Collaborator Guide — Connecting Your Computer to the Study

Welcome! This guide takes you from zero to done. **No coding and no typing of commands is required** — you'll install one free app, put your data in a folder, copy a code from the website, and **double-click one file**. That's it.

Total time: about **10–15 minutes**, once.

What this is (in plain words)

The study uses a shared website to coordinate research. The heavy analysis on your data must run **on your own computer**, so your **raw data never leaves your machine**. To do that, you run one small background helper — the **Site Agent**. It quietly waits for tasks from the study, runs them locally, and sends back **only summary results** (never your raw data).

The privacy promise

- ✓ Your raw data stays on your computer. The helper opens it **read-only** — it can't change or delete your files.
- ✓ Only safe summaries (filtered marker lists, counts, population coordinates) go to the study server.
- ✓ The helper only makes **outgoing** connections. Nothing connects *into* your computer — no firewall changes needed.

The whole journey at a glance

1. Install **Docker Desktop** (free app) — once.
2. Put your data in a folder (a simple layout, shown below).
3. On the **website**: log in → register your dataset's details (phenotype name + number of samples) → copy your one-time **connection code**.
4. On **your computer**: **double-click "Start Agent"** and answer 3 short questions. It connects **once** and then stays ready in the background.
5. **Join or start collaborations anytime** — the already-running helper picks up your tasks automatically (even ones created while your computer was off).

You set up steps 1–4 **once**. After that you only ever do step 5, and the helper handles the rest by itself.

> *The single most important link to get right: the **phenotype name you type on the website must exactly match the data sub-folder name on your computer** (e.g. website phenotype `eye_color` ↔ folder `eye_color/rawdata.csv`). That's how the helper knows which file a task is about.*

Before you start — checklist

- A Windows or Mac computer you can leave on sometimes.
- Your genotype data saved as a **CSV** file.
- The **study link** (your coordinator gives you this — e.g. `https://3-21-x-x.sslip.io`). This one link is **both** the website you log into **and** the "server address" you'll paste into the helper. (It may look like a string of numbers + `.sslip.io` instead of a normal name — that's expected and fine.)
- The **SiteAgent kit** from your coordinator — usually a `.zip`. Save it somewhere easy and **unzip it**; inside is the "Start Agent" file you'll double-click and this guide. You don't open or edit anything inside it.

Step 1 — Install Docker Desktop (free, one time)

Docker Desktop is the engine that runs the helper safely in the background.

Mac

1. Go to https://www.docker.com/products/docker-desktop/ → download for Mac. (Not sure which chip? Apple menu → *About This Mac*. "Apple M1/M2/M3" = Apple chip; otherwise Intel.)
2. Open the downloaded file and drag **Docker** into **Applications**.
3. Open **Docker** from Applications (it may ask for your password the first time).
4. Wait until the whale icon in the top menu bar is steady and says **"running"**.

Windows

1. Go to https://www.docker.com/products/docker-desktop/ → download for Windows.
2. Run the installer → **Next → Next → Finish** (accept defaults). Restart if asked.
3. Open **Docker Desktop** from the Start menu. Wait until the bottom-left corner is **green / "running"**.

> *Do Step 1 only once. Leave Docker Desktop set to start with your computer (it does by default), so the helper can run whenever your machine is on.*

Step 2 — Put your data in a folder

Make one **top folder** (anywhere easy), and inside it create **one sub-folder per dataset**, named exactly like the **phenotype you register on the website**. Each sub-folder holds a file called `rawdata.csv`.

```
collab-data/                    ← your top folder (any name/location)
├── eye_color/                  ← sub-folder name = the phenotype you register
│   └── rawdata.csv
└── blood_pressure/
    └── rawdata.csv
```

Telling the system who is a case vs a control — pick whichever is easier:

- **Option A:** include a column named `phenotype` in `rawdata.csv` (**1 = case, 0 = control**), or
- **Option B:** add two text files next to `rawdata.csv` — `case_ids.txt` and `control_ids.txt` — each listing one sample ID per line.

```
blood_pressure/                   eye_color/
└── rawdata.csv (phenotype col)   ├── rawdata.csv (no phenotype column)
                                  ├── case_ids.txt
                                  └── control_ids.txt
```

The data standard (please follow this exactly)

So everyone's results line up. **If your file came from your lab's standard genotype export, this is usually already true** — when unsure, ask whoever prepared it.

Rule	What it means
CSV file named `rawdata.csv`	Plain comma-separated file.
First column = Sample ID	One row per person; IDs **unique**; same IDs used in the case/control files if you use those.
Other columns = markers (SNPs)	One column per marker.
Column names = marker IDs (e.g. `rs12345`)	Use the **same naming the study agreed on**; the same marker has the same name at every site.
Values = 0, 1, or 2	Copies of the marker's alternate allele. **Leave blank for missing** (don't use 0 for missing — 0 is a real value).
Same export style across the study	So a `0/1/2` means the *same thing* everywhere (same genome build / pipeline, e.g. GRCh38). Your bioinformatician can confirm in one line.
Case/control	A `phenotype` column (1/0) **or** `case_ids.txt` + `control_ids.txt`. Needed for GWAS.

Tiny example of `rawdata.csv`:

```
sample_id,phenotype,rs12345,rs67890,rs11223
NA2000,1,0,1,2
NA2001,0,1,1,0
NA2002,1,,2,1        ← blank = missing genotype
```

How your data is handled (so there are no surprises)

You **don't** need the same markers as other sites — the system reconciles differences automatically:

- **Markers not in the study's shared reference are skipped for the population-similarity step only** (Population Stratification) — they're still used in your

QC checks and in GWAS, so nothing is wasted.

• **Missing some reference markers is fine** — they're filled in neutrally. **Extra** markers are fine too — just skipped for that one step.

• **Sharing almost none of the reference markers** → the population step is skipped for your dataset (expected, not an error); everything else still runs.

• **Two sites with different markers still work** — each is lined up against the shared reference independently, so results stay comparable.

• **Your file is fingerprinted** when registered — if you edit it later, re-register it on the website.

Step 3 — On the website

Everything in this step is done in your **web browser** on the study website. (The website is just for coordinating — your actual genetic data never goes here.)

3a. Create an account / log in

1. Open the study website address your coordinator gave you.
2. **Register** (name, email, password) or **Log in** if you already have an account.

3b. Register your dataset's details (metadata only)

This tells the study *that* you have a dataset and *how big* it is — **not the data itself.** The data stays on your computer.

1. Go to the **Upload / "Enter Metadata Details"** page.
2. Fill in two fields:

• **Phenotype(s):** the dataset name. **Type it exactly the same as your local sub-folder** (e.g. `eye_color`). This is the link between the website and your computer.

• **# of Samples:** how many individuals are in that dataset (e.g. `20`).

3. Click **Submit.** You'll see "Metadata uploaded successfully."

No file is uploaded here — only the name and the sample count. Repeat 3b for each dataset you have (one per phenotype/folder).

3c. Get your connection code

This is a **one-time login for the helper** — it lives on your **Edit Profile** page. **Do this before joining any collaboration**, so your helper is ready and waiting. It's a little tucked away, so follow these clicks exactly:

1. Look at the **top-right corner** of the website and click your **account icon** (the round avatar / circle with your initials or a person symbol).
2. A small menu drops down. Click **"Edit Profile."**
3. The Edit Profile page opens. **Scroll down** — past the Name, Email, and Password boxes — to a box titled **"Connect my computer."**

4. In that box, click the **"Generate connection code"** button.
5. A long code appears just below the button. Click **"Copy code"** (the button next to it). It's a long string of random-looking letters and numbers — that's normal, and you don't need to read or understand it.

The code is now on your clipboard — go straight to Step 4 and paste it when asked. (If you wait too long, see the note below and just generate a fresh one.)

> ***You only ever need a code once.*** *The code itself expires in **24 hours**, but as soon as the helper uses it, the helper saves a permanent login — so you **never need a new code** for new collaborations or after restarting your computer. You'd only generate another one if you completely reset the helper, or if the code expired before you used it (just repeat 3c to get a fresh one).*

Step 4 — Start the agent (double-click, no typing)

Open the **SiteAgent folder** your coordinator gave you.

Mac — double-click `Start Agent.command`.

- First time only, macOS may warn it's from an unidentified developer. If so: **right-click** the file → **Open** → **Open**. (You only do this once.)

Windows — double-click `Start Agent.bat`.

- If Windows SmartScreen warns, click **More info → Run anyway** (one time).

A window opens and asks **three short questions**:

1. **Study server address** — paste the website address from your coordinator.
2. **Your top data folder** — the folder from Step 2 (e.g. `collab-data`).
3. **Your connection code** — paste the code you copied in Step 3.

The helper then checks Docker, sets everything up, and starts. When you see **"Agent started"**, you're done with setup. You can close the window — it keeps running in the background and restarts automatically when you turn your computer on.

Your helper is now **enrolled once and ready for everything** — you won't touch any of Steps 1–4 again.

Step 5 — Join or start collaborations (anytime)

With the helper already running and ready, this is all you do from now on. It can happen **before or after** the helper started — order doesn't matter, because tasks wait until your helper is online.

- **If you were invited to a study:** on the website open **Invitations** (or the Collaborations page), find the study, and click **Accept** (pick your matching dataset/phenotype if asked).

- **If *you* are starting a study (initiator):** open **Start Collaboration**, give it a name, pick the **GWAS** experiment, choose the **QC scheme** (your coordinator says which — e.g. MAF / HWE / Missing; add Population Stratification only if your data matches the study's reference panel), select **your dataset**, and **invite** collaborators by email.

Once everyone invited has accepted, the study automatically queues the analysis tasks for each person's computer. Your already-running helper **picks up your task on its own** — nothing else for you to do. Watch progress on the website (or with the optional `logs` command below).

> *No new connection code is needed for each collaboration — your helper is already logged in. Just accept/create studies and it handles them.*

Step 6 — Run the analysis and view results

This all happens **on the website**, and is mostly done by the **study initiator** (the person who started the collaboration). Invited collaborators usually just keep their helper running and watch the results appear. **The page updates by itself — you do not need to refresh your browser** after clicking these buttons; the next stage appears automatically once everyone's helper has finished its part.

Open the collaboration's **Details** page. The **Collaboration Status** panel on the right shows which stage you're in. The buttons appear as each stage becomes ready:

1. **QC runs automatically.** When collaborators accept, each one's helper runs the QC locally and uploads its results. You'll see the status move past "Onboarding."
   - For filter-only schemes (MAF / HWE / Missing), it goes straight to the GWAS stage — there's no extra QC button to press.
   - For schemes with **Population Stratification** or **Sample Relatedness**, the initiator clicks **"Initiate QC Calculation,"** then **"Get QC Results,"** and sets a cutoff with **"Confirm Threshold."**
2. **Create the GWAS dataset.** Click **"Create GWAS Dataset."** Each helper computes its per-marker case/control counts locally and uploads them. (Keep helpers running.) The status advances on its own when they're done.
3. **Run the analysis.** Click **"Initiate GWAS Calculation."** The server combines everyone's counts and runs the statistics.
4. **See the results.** Click **"Get GWAS Results"** to view the results table. Optionally click **"Generate Summary"** for a plain-language one-page summary.

> *If a button says results aren't ready yet, just wait a few seconds — the page refreshes itself as each collaborator's helper reports in. You don't need to reload.*

Day-to-day — what to expect

- **You don't have to do anything.** The helper waits for tasks and handles them.

- **You don't need the website open** or to stay logged in — the helper has its own saved login.
- **Tasks can arrive anytime.** If your computer was off or asleep when a task was created, no problem — it's picked up the next time you're on and online.
- **Results appear on the website** for the study team once your helper finishes.

If you ever need to check or control it (optional)

Easiest: just **double-click "Start Agent" again** — it will make sure the helper is running. For more control, open the SiteAgent folder in Terminal (Mac) / PowerShell (Windows) and use:

What you want	Mac	Windows
Is it running?	`./collab-agent.sh status`	`.\collab-agent.ps1 status`
Watch it work	`./collab-agent.sh logs`	`.\collab-agent.ps1 logs`
Stop it	`./collab-agent.sh stop`	`.\collab-agent.ps1 stop`
Update it	`./collab-agent.sh update`	`.\collab-agent.ps1 update`
Start over	`./collab-agent.sh reset`	`.\collab-agent.ps1 reset`

A healthy helper logs `Agent running. Polling for jobs ...` and, when work arrives, `Job ... complete; uploaded keys=[...]`.

Troubleshooting

"Docker is not installed / not running" Open **Docker Desktop** and wait until it says *running* (steady whale / green), then double-click "Start Agent" again.

Mac: "Start Agent" won't open (unidentified developer) Right-click the file → **Open** → **Open**. You only need to do this the first time.

Windows: SmartScreen blocked it Click **More info → Run anyway** (one time).

"No dataset for phenotype 'xxx'" The task needs a dataset you don't have locally. Check your top data folder has a sub-folder named exactly `xxx` with a `rawdata.csv` inside.

"...does not match the registered hash" Your `rawdata.csv` changed after you registered it. Re-register the dataset on the website (or restore the original file), then it runs.

"Invalid or expired connection code" Codes expire after 24 hours. Generate a fresh one (Step 3), then double-click "Start Agent" again and paste the new code.

It can't reach the server Check your internet and that the server address is right. The helper keeps retrying, so it resumes on its own once you're back online.

Frequently asked questions

Does my raw genetic data ever get uploaded? No. Raw files are read on your computer only (read-only). Only summarized, privacy-safe outputs are sent.

Do I need a new connection code for each collaboration, or each time I restart? No — just **once, ever.** The helper turns your one-time code into a saved login that covers **all** your collaborations and survives restarts and computer reboots. You'd only need a fresh code if you fully **reset** the helper.

Should I set up the helper before or after joining a study? Either works, but **setting it up first is best** — then it's ready and waiting. If a task is created while your helper is off, it simply waits and runs when you're next online.

Do I have to keep a window open? No. It runs in the background — you can close the window after it starts.

What if I turn off or restart my computer? It restarts automatically with your computer (as long as Docker Desktop is running). Tasks that arrived while you were off are picked up afterward.

Do I need the same markers (SNPs) as other sites? No. You can have more or fewer — the system lines each site up against a shared reference automatically. Just follow the data standard above.

Why does it say some of my markers weren't used? For the population-similarity step, only markers in the study's shared reference can be used (everyone is compared on the same "ruler"). Extras are skipped *for that step only* — still used in your QC and GWAS, so nothing is wasted.

My dataset has fewer markers than the reference — is that OK? Yes, as long as you share a good chunk of them (missing ones are filled in neutrally). If you share almost none, that one step is skipped for you — expected.

Will this slow down my computer? Only briefly while it's actively working on a task, and tasks are occasional.

How do I remove it completely? In the SiteAgent folder run `./collab-agent.sh reset` (Mac) or `.\collab-agent.ps1 reset` (Windows). You can then quit/uninstall Docker Desktop if you like.

Who do I contact for help? Your study coordinator. Sending them the output of the **logs** command (above) helps them help you fast.